



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
 December 2024

EE2003-1 – MICROCONTROLLER
 (For EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- Auxiliary carry is set during which condition?**
 - When carry is generated from D3 to D4
 - When carry is generated from both D3 to D4 and D7
 - When carry is generated from D7
 - When carry is generated at either D3 to D4 or D7
- TQFP stands for**
 - Thin Quad Flat Package
 - Thick Quad Flat Package
 - Thick Quad Flip Package
 - Thin Quad Flip Package
- ACC= (0x45) | (0x0F). Output is**
 - 0x04
 - 0x05
 - 0x4F
 - 0x45
- The port which has no alternate function other than bidirectional input-output is**
 - Port 0
 - Port 2
 - Port 1
 - Port 3
- What is the operation for Timer in mode 2?**
 - 13-bit timer mode, 8-bit timer/counter THx and TLx as 5-bit prescaler
 - 8-bit auto reload mode, 8-bit auto reload time/counter; THx holds a value which is to be reloaded into TLx each time it overflows
 - 16-bit timer mode, 16-bit timer/counter THx and TLx are cascaded, no prescaler
 - Split timer mode
- To enable external interrupt 0, the value to be loaded to IE register is**
 - 0x81
 - 0x01
 - 0x04
 - 0x84
- In mode 2, timer 1 register overflows when the register reaches**
 - 0x0F
 - 0xFFFF
 - 0x0FFF
 - 0xFF
- If microcontroller is operated with 12 MHz crystal frequency, then the timer clock frequency is _____ MHz.**
 - 1us
 - 10us
 - 1.085us
 - 10.85us
- _____ port pin is used for transmission of serial data.**
 - P3.1
 - P3.0
 - P3.2
 - P3.3
- PS bit in _____ register must be set to give the serial data interrupt highest priority**
 - IE
 - IP
 - SCON
 - PCON
- Baud rate is defined as**
 - Bytes transmitted in one second
 - Bits transmitted in one minute
 - Bytes transmitted in one minute
 - Bits transmitted in one second

12. In RS232 serial I/O interfacing standard, the logic 1 is represented using
 A) 0 to 5V B) -5V to +5V
 C) -3V to -25V D) +3V to +25V
13. _____ bit ADC provides best resolution
 A) 4 B) 8
 C) 16 D) 12
14. _____ logic is given to turn ON common cathode LEDs
 A) 0 B) 1
 C) Either 1 or 0 D) Depends on circuit
15. In PWM technique, speed control of DC motor, _____ is true
 A) Wider pulse, higher speed B) Wider pulse, lower speed
 C) Independent of pulse width D) None of these
16. 8 input DAC has _____
 A) 128 discrete voltage levels B) 8 discrete voltage levels
 C) 64 discrete voltage levels D) 256 discrete voltage levels
17. HC-05 is a _____ module
 A) Bluetooth Module B) IR sensor module
 C) PIR sensor module D) Temperature sensor module
18. Role of diode connected across magnetizing coil of relay
 A) To increase the current flow through the coil B) To protect the circuit from voltage spikes
 C) To reduce power consumption D) To provide additional magnetic force
19. Acronym of PIR sensor
 A) Profile Infrared Sensor B) Positive Infrared Sensor
 C) Passive Infrared Sensor D) Photo Infrared Sensor
20. LED is always provided with series resistor
 A) To increase the brightness of the LED B) To improve the LED's colour
 C) To reduce power consumption D) To limit the current flowing through the LED

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Explain the program memory organization of SST89X516RD2 microcontroller. | 4 | L*1 | 1 | 1 |
| b) Write a program to perform any two logical operations on two 8-bit numbers. Show the result in internal memory of microcontroller | 4 | L3 | 1 | 1 |
| c) Explain the method and steps to program a timer in mode2 configuration | 8 | L2 | 2 | 2 |
| 2. a) Explain the bit fields of IP and IE registers | 4 | L1 | 2 | 2 |
| b) Identify the addressing modes | | | | |
| i. ACC = *internalmemory | | | | |
| ii. *R1 = 0x40 | | | | |
| iii. *R7 = ACC | | | | |
| iv. *internalmemory = ACC | 4 | L1 | 1 | 1 |
| c) Write a program to toggle P2.0 at 1kHz rate with 50% duty cycle. Use Timer1 in mode1. Assume f= 12MHz. | 8 | L3 | 2 | 2 |
| 3. a) List any 4 salient features of SST89X516RD2 | 4 | L1 | 1 | 1 |
| b) Draw the internal structure of timer | 4 | L2 | 2 | 2 |
| c) Write a C program that continuously gets a single bit of data from P1.7 and sends it to P1.0, while simultaneously creating a square wave of 200µs period on pin P2.5. Use Timer 0 in mode 2 to create the square wave. Assume that XTAL = 11.0592 MHz. | 8 | L2 | 2 | 2 |

Unit – II

- | | | | | |
|--|---|----|---|---|
| 4. a) Explain the steps to program the microcontroller to transfer data serially. Also tell the importance of TI flag while transmitting data. | 8 | L1 | 3 | 3 |
|--|---|----|---|---|

- | | | | | | |
|-------------------|---|---|----|---|-------------------|
| b) | Write a program to read analog signal using ADC connected P0. Display the digital value in LED array connected to P2. The control lines of ADC start, OE, EOC is connected to P3.6, P3.5 and P3.7 respectively. | 8 | L3 | 4 | 1,2,3,5,
9,10 |
| 5. a) | Write a program to display up counter 0xAA to 0xBB in a multiplexed seven segment display. The data lines are connected to P1. The control pins are connected to P3.6 and P3.7 | 8 | L3 | 4 | 1,2,3, 5,
9,10 |
| b) | Explain the bit fields of SCON and PCON registers used in serial communication. | 8 | L2 | 3 | 3 |
| 6. a) | Write a program to blink LED array connected to P2 at 1Hz 5 times. | 8 | L4 | 4 | 1,2,3,5,
9,10 |
| b) | Write a program to send "NMAMIT" serially at 9600 baud rate continuously. | 8 | L2 | 3 | 3 |
| Unit – III | | | | | |
| 7. a) | Design a microcontroller based system to maintain the temperature in a room at 20 degree Celsius. | 8 | L4 | 5 | 1,2,3,5,
9,10 |
| b) | Moisture level in an agricultural farm need to maintained at 60%. Design a suitable system using microcontroller and suitable sensors. | 8 | L4 | 5 | 1,2,3,5,
9,10 |
| 8. a) | Design a system to monitor water level of a tank in real time. Display the water level in 16x2 LCD along with buzzer indication at 10% and 90% water level indications. | 8 | L4 | 5 | 1,2,3,5,
9,10 |
| b) | It is required to measure voltage across a 24V battery. Design a suitable system using microcontroller and display the voltage in LCD. | 8 | L4 | 5 | 1,2,3,5,
9,10 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
